

Outlier removal using the Box-Cox transform and then the Z-score method

As a civil engineer, you are analyzing the dataset, and believe that there are outliers in compressive strength, which need to be removed before the dataset can be used for model training.

Please identify these outliers using the **Box-Cox transform and then the Z-score method**.

* The compressive strength values can be imported using the code below.

```
T = readtable("Concrete_Data.xls", 'VariableNamingRule','preserve');
values = table2array(T(:,9));
```

Script

 Save  Reset  MATLAB Documentation (https://www.mathworks.com/help/) [Open Item in MATLAB Online](#) 

```
12
13 [X_temp, lambda] = boxcox(X);
14 % Draw the histogram of the transformed distribution
15 histogram(X_temp);
16
17 % Calculating the z-score of every sample in the dataset. Your code:
18 z = (X_temp - mean(X_temp)) ./ std(X_temp);
19 % Plotting the distribution of z
20 figure
21 histogram(z)
22
23 % Filtering out those samples having large z-scores
24 % Removing outliers, z should be between -3 and 3. Your code:
25 X = X_temp(abs(z) < 2.5);
26
27 histogram(X)
28
```

 Run Script 

Tests: All Tests Passed (100%)

Submit 

 Usage of boxcox	20% (20%)
 Keep samples that are not outliers	20% (20%)
 Outlier test #1	20% (20%)
 Outlier test #2	20% (20%)
 Outlier test #3	20% (20%)

Total: 100%

Output

